WORKED SOLUTION

STEP III 2020, Question 10

Mechanics • Simple Harmonic Motion

STEP III 2020, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Hanging from a spring, the particle obeys simple harmonic motion about its equilibrium position, and that fixes Ω in terms of k, g and a. On the platform, the particle's acceleration is the platform's, which is known in terms of x, so Newton's second law gives the reaction directly. Contact needs the reaction to be non-negative all the time; since it turns out to be *linear* in x, we only have to check it at the two ends of the platform's travel, and which end matters depends on whether ω is smaller or larger than Ω.

Solution

Model the particle as a point mass and the spring as light, obeying Hooke's law in both extension and compression. Throughout, take upwards as positive for forces on the particle.

The hanging particle. Let y be the distance of the particle below A, so the extension is $y - a$ and the tension is $kmg(y-a)/a$, acting upwards on the particle. Newton's second law, measured downwards, gives

$$m\ddot{y} = mg - \frac{kmg}{a}(y-a) \qquad \Longrightarrow \qquad \ddot{y} = -\frac{kg}{a}\left(y - a - \frac{a}{k}\right).$$

With $z = y - a(1+\frac{1}{k})$ this is $\ddot{z} = -\frac{kg}{a}z$, which is simple harmonic motion about the equilibrium point $y = a(1+\frac{1}{k})$. The particle is released from rest away from this point (any extension other than a/k gives $z \neq 0$), so it oscillates vertically with angular frequency $\sqrt{kg/a}$ and period $2\pi\sqrt{a/(kg)}$. Setting this equal to $2\pi/\Omega$ gives $\Omega^2 = kg/a$, that is,

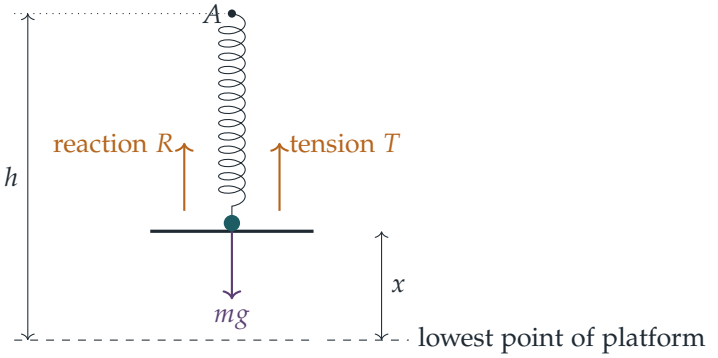
$$kg = a\Omega^2.$$

We use this in the form $g/\Omega^2 = a/k$ below.

The platform. The platform performs simple harmonic motion of amplitude b and angular frequency ω, so its height x above its lowest point ranges over $0 \le x \le 2b$, with centre $x = b$. Hence its acceleration is

$$\ddot{x} = -\omega^2(x - b) = \omega^2(b - x).$$

(For instance, $x = b(1 - \cos\omega t)$ gives $\ddot{x} = \omega^2 b\cos\omega t = \omega^2(b - x)$.)



The reaction. While the particle is in contact, it moves with the platform, so its height above the platform's lowest point is x and its acceleration is $\ddot{x}$. Its distance below A is $h - x$, so the extension of the spring is $h - x - a$ and the tension is

$$T = \frac{kmg}{a}(h - x - a) = m\Omega^2(h - x - a),$$

using $kg/a = \Omega^2$. (If $h - x < a$ the spring is compressed and T is negative, a thrust, and the same expression still holds.) The forces on the particle are R and T upwards and mg downwards, so

$$R + T - mg = m\ddot{x} = m\omega^2(b - x),$$

and therefore

$$R = mg - m\Omega^2(h - x - a) + m\omega^2(b - x) = mg + m\Omega^2(a + x - h) + m\omega^2(b - x),$$

as required.

Staying in contact. The particle stays on the platform throughout exactly when $R \ge 0$ for every x in $[0, 2b]$. Collecting the terms in x,

$$\frac{R}{m} = g + \Omega^2(a - h) + \omega^2 b + (\Omega^2 - \omega^2)\,x.$$

? **Ask yourself** Why is it enough to look at only one value of x?

R is a linear function of x, so over the interval $0 \le x \le 2b$ its least value is at one end: at $x = 0$ if the slope $\Omega^2 - \omega^2$ is positive, and at $x = 2b$ if it is negative. The platform reaches both ends in every cycle, so $R \ge 0$ throughout is equivalent to $R \ge 0$ at the appropriate end.

Case $\omega < \Omega$. The least value is at $x = 0$ (the bottom of the motion), so we need

$$g + \Omega^2(a - h) + \omega^2 b \ge 0 \iff h \le a + \frac{g}{\Omega^2} + \frac{\omega^2 b}{\Omega^2} = a\Big(1 + \frac{1}{k}\Big) + \frac{\omega^2 b}{\Omega^2},$$

dividing by $\Omega^2 > 0$ and using $g/\Omega^2 = a/k$. This is the required inequality.

Case $\omega > \Omega$. Now the least value is at $x = 2b$ (the top of the motion), so we need

$$g + \Omega^2(a - h) + \omega^2 b + 2b(\Omega^2 - \omega^2) \ge 0 \iff h \le a\Big(1 + \frac{1}{k}\Big) + b\Big(2 - \frac{\omega^2}{\Omega^2}\Big).$$

A bound for every ω. Suppose the particle stays in contact.

- If $\omega < \Omega$, then $\omega^2 b/\Omega^2 < b$ (as $b > 0$), so $h \le a(1 + \frac{1}{k}) + \omega^2 b/\Omega^2 < a(1 + \frac{1}{k}) + b$.
- If $\omega > \Omega$, then $2 - \omega^2/\Omega^2 < 1$, so $h \le a(1 + \frac{1}{k}) + b(2 - \omega^2/\Omega^2) < a(1 + \frac{1}{k}) + b$.
- If $\omega = \Omega$, the x terms in R cancel and $R/m = g + \Omega^2(a - h) + \Omega^2 b$ is constant; $R \ge 0$ gives exactly $h \le a(1 + \frac{1}{k}) + b$.

So in every case

$$h \le a\Big(1 + \frac{1}{k}\Big) + b$$

is necessary, whatever the value of ω. Since $h - b$ is the depth of the centre of the platform's motion below A, and $a(1 + \frac{1}{k})$ is the depth of the hanging equilibrium, this says: the platform must oscillate about a point no lower than the particle's natural equilibrium position.

Checks

- Stepping the free motion numerically with $a = 1$, $k = 2$, $g = 9.8$ gives a period with $a\Omega^2 \approx 19.600$, matching $kg = 19.6$.
- If the platform is at rest ($b = 0$), both bounds become $h \le a(1 + \frac{1}{k})$: the platform must not be below the equilibrium point, or the spring would lift the particle off. That is exactly right.
- At $\omega = \Omega$ the two case bounds agree, both giving $a(1 + \frac{1}{k}) + b$, as they should by continuity.

Common slips

- Checking only the top of the motion. For slow platforms ($\omega < \Omega$) the particle is most at risk at the *bottom*, where the spring pulls up hardest.

- Forgetting that the spring can be compressed: the tension formula $kmg(h-x-a)/a$ is valid with either sign, so no separate case is needed.

Beyond STEP

This is a forced oscillator in disguise: the spring tries to make the particle oscillate at its natural frequency Ω, while the platform imposes frequency ω, and the reaction is the force needed to reconcile the two. Undergraduate courses on oscillations study driven systems, resonance and the way the response depends on ω/Ω. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats these ideas carefully.

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